

Vikram Raj Nagoor Kani

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EDUCATION

University of Illinois Urbana Champaign

M.Eng in Autonomy and Robotics

Urbana Champaign, IL
Aug 2024 - Expected May 2026

BS Abdur Rahman Crescent Institute of Technology

B.Tech in Electronics and Communication Engineering

India
Jul 2017 - Jun 2021

WORK EXPERIENCE

Founding Robotics Engineer, Contract, Premier Cooperative

Aug 2025 - present

- Designed a low cost solution to digitally override a joystick control box of a grain probe using a custom **PCB**, **ESP32** microcontroller and **NVIDIA Jetson** enabling automation of grain sample collection from a truck.
- Developed and containerized a **ROS 2** pipeline using **Docker** that detects incoming grain trucks using **YOLO**, builds a spatial map using Intel Depth Sense camera and controls the grain probe to take grain samples every 45 seconds.
- Building a web based control interface using **React**, **FastAPI** and **Uvicorn** to configure the system, calibrate cameras, override operations, and start/stop the robotic pipeline.

Robotics Perception Engineer, Apprentice, Kohler

Jun 2025 - Aug 2025

- Developed a real time product packaging assistant and validation system for faucets using a lightweight **Mask RCNN** instance segmentation model, achieving **30 ms inference latency** on **NVIDIA Jetson**.
- Generated a synthetic dataset to augment missing faucet color and finish variations, resulting in a model achieving above **90% accuracy** and **mAP** on real world inspection.

Artificial Intelligence Engineer, Intern, UIUC College of Veterinary Medicine

Jun 2025 - Aug 2025

- Worked as a research assistant for **Dr. Ben Blair**, building AI training pipelines for swine wellness research.
- Evaluated state of the art pose detection models trained on homogeneous and heterogeneous swine datasets to measure their impact on keypoint detection accuracy in real world farm environments.
- Developed a web based application using **React** and **Uvicorn** to detect swine wellness through fecal matter classification.

Robotics Perception Engineer, Intern, UIUC College of Applied Health Sciences

Jan 2025 - May 2025

- Optimized multimodal deep learning **ROS2** pipeline for STRETCH Robot AI, enabling compatibility with **low end GPUs** by reducing VRAM usage from **12GB to 7GB**.
- Improved the reliability and safety of elderly assistive object pick and place tasks, **doubling** the pickup **success** rate from **20% to 40%**.

Automation Engineer, Fulltime, Accenture

Aug 2021 - Jul 2024

- Designed an automated system to monitor and maintain **Honeywell Lenel S2** security systems deployed in Microsoft Datacenters globally, ensuring badge readers, CCTV cameras, and security applications operated reliably, and triggering alerts when an edge device fails.
- Evaluated performance of **cognitive vision models** by assessing accuracy in detecting and reporting faulty surveillance cameras across Microsoft data centers.
- Developed **Azure CI/CD pipelines** to provision, configure, and manage **Docker** containers, **Azure** virtual machines, and **BareMetal** servers, reducing manual setup and maintenance effort by **60%**.

Embedded System Engineer, Intern, Nokia

Feb 2021 - May 2021

- Developed an **IoT device** to detect obstacles blocking access to fire extinguishers and alert security in **real time** for 5G Radio Chip Manufacturing Factory.
- Built a **Python Flask** backend with **SQL** database to communicate with **ESP32** microcontroller to continuously update sensor data and transmit real time status updates to the security system.

OPEN SOURCE CONTRIBUTION

ROS2 Multi Stream RTSP Image Bridge

ROS2, C++, OpenCV

- Added **multi stream support** enabling transmission of multiple ROS2 image topics as RTSP streams.
- Implemented **authentication** with custom username and password for creating secure RTSP streams.

Autonomous Lane Tracking and Control for F1TENTH Racecar 🔗

ROS, Python, Open CV, NVIDIA Jetson, F1 Tenth, Unity, Linux

- Developed a lightweight lane detection algorithm using pure **OpenCV**, improving computational **speed by 30%** on **NVIDIA Jetson** for real time waypoint generation; accepted as an **open source contribution** for UIUC ECE 484 🔗.
- Implemented sensor fallback between **camera** and **LiDAR** based waypoint generation for robust operation in low visibility conditions and fine tuned a **PID controller** to follow the waypoints using **Ackermann steering**.
- Built a **photorealistic digital twin** of the **F1TENTH** racetrack and vehicle using a **Unity based simulator**, enabling rapid testing and validation with **minimal sim to real gap**.

Persistent Pedestrian Detection and Tracking for Autonomous Vehicles 🔗

GEM E4, PyTorch, ROS, Python, Open CV, YOLO, GEMStack

- Developed a persistent pedestrian detection and tracking pipeline for the **GEM e4 autonomous vehicle** using **camera** and **LiDAR** data. 2D pedestrian detections from **YOLO** were projected into **3D bounding volumes** using LiDAR to build a spatial world model for robust tracking across frames.
- Optimized the perception pipeline for **real time performance** using techniques such as **voxel downsampling**, reducing LiDAR processing overhead.

Photorealistic Simulation Environment for Autonomous Vehicles 🔗

Gazebo, ROS, Python, GEMStack, MAST3R-SLAM, OpenCV

- Built a **photorealistic 3D digital twin** of the UIUC Highbay Lab using **MASt3R-SLAM** to reconstruct dense meshes from real world scans and integrated the environment into **Gazebo** for simulation of **GEM e2/e4 autonomous vehicles**.
- Developed a **text-based scene description interface** enabling dynamic spawning and manipulation of objects within the simulation during runtime to support testing of perception and navigation algorithms.

6-DoF Object Pose Estimation for Robotic Arm Manipulation 🔗

PyTorch, Python, OpenCV, ResNet, YOLO

- Designed a **deep learning pipeline** to estimate **6-DoF object pose** for robotic manipulation using **RGB-D inputs**.
- Implemented a **dual-stream U-Net architecture** with hybrid **feature fusion** to combine RGB and depth features for accurate pose prediction.
- Built a **custom data loader** and augmentation pipeline to convert **BOP dataset** annotations into **YOLO** compatible format, enabling seamless training and experimentation with detection based pose estimation models.

TECHNICAL SKILLS

Programming Languages : Python, C++, Bash, PowerShell, LabView, Matlab, Azure CLI.

Technologies/Frameworks : Pytorch, Open CV, ROS, ROS2, Docker, RoboFlow, Anaconda , NVIDIA Jetson, GEM E2, GEM E4, F1Tenth, Azure, Gazebo, Gazebo Ignition, Unity, Git, Linux, FreeCAD, Open AI Gym, Deep Learning, Machine Learning, Transformers, UR3e, CI/CD, easyEDA, React, Unicorn, SQL, MongoDB, Python Flask, Fast API.

Coursework : Autonomous Vehicle System Engineering, Computer Vision, Deep Learning with Computer Vision.